MIL-F-18180/3A
30 May 1986
SUPERSEDING
MIL-F-18180/3
30 April 1981

MILITARY SPECIFICATION SHEET

FLANGES AND FLANGED FITTINGS, PIPE, STEEL, CLASSES 150, 300, 400, 600, 900, 1500, AND 2500; SLIP-ON WELDING FLANGES

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the slip-on welding flanges described herein shall consist of this document and the latest issue of specification MIL-F-18180.



ANSI B16.5



CONVOLUTED

FIGURE 1. Slip-on welding flanges.

1. Military specification sheet part number. Slip-on welding flanges covered by this specification sheet shall be designated in the following form (see 6.2(b) of MIL-F-18180).

M18180/3 XXX X XX X

Military Specification Sheet Code Number

Pipe Size and Pressure-Temperature Rating Class Code Number (see para 2)

Flange Facing Code Letter (see para 3)

ANSI B16.5 Materials Group No. Code Number (see para 4)

Reduced Size Code Letter (for Reducing Slip-on Flanges Only (see para 5)

FSC 4730

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2. Pipe size and pressure-temperature rating. The nominal pipe size and pressure-temperature rating class of slip-on welding flanges are designated by two digits (see table I).

TABLE I. Code number to size and class.

Nominal Pipe Size (inch)	Pressure-Temperature Ratings Class 150	Class 300	Class 400	Class 600	Class 900	Class 1500
1/2	01	21	Use code No. for same size in class 600	52	Use code No. for same size in class 600	84
3/4	02	22		53		85
1	03	23		54		86
1-1/4	04	24		55		87
1-1/2	05	25		56		88
2	06	26		57		89
2-1/2	07	27		58		90
3	08	28		59	72	--
3-1/2	09	29		60	--	--
4	10	30	41	61	73	--
5	11	31	42	62	74	--
6	12	32	43	63	75	--
8	13	33	44	64	76	--
10	14	34	45	65	77	--
12	15	35	46	66	78	--
14	16	36	47	67	79	--
16	17	37	48	68	80	--
18	18	38	49	69	81	--
20	19	39	50	70	82	--
24	20	40	51	71	83	--

3. Flange facing. The facing of slip-on flanges is designated by a single letter (see table II).

TABLE II. Code letter to flange facing.

Code Letter	Flange Facing
A	Raised
B	Flat
C	Ring-Joint
D	Large Male
E	Large Female
F	Small Male
G	Small Female
H	Large Tongue
J	Small Tongue
K	Large Groove
M	Small Groove

4. ANSI B16.5 material group number. The ANSI B16.5 Material Group Number for slip-on welding flanges is designated by two digits (see table III).

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TABLE III. Code number to material group number.

Code Number	ANSI B16.5 Material Group No.	ANSI B16.5 Acceptable Material (ASTM Standards for)
01	Group 1.1	A105, A216 Grade WCB, A350 Grade LF2, A515 Grade 70, A516 Grade 70 or A537 Class 1.
02	Group 1.2	A203 Class B or E, A216 Grade WCC, A350 Grade LF3 or A352 Grade LC2 or LC3.
04	Group 1.4	A350 Grade LF1, A515 Grade 60 or A516 Grade 60.
05	Group 1.5	A182 Grade F1, A204 Grade A or B, A217 Grade WC1 or A352 Grade LC1.
07	Group 1.7	A182 Grade F2, A204 Grade C or A217 Grade WC 4 or WC 5.
09	Group 1.9	A182 Grade F11 or F12, A217 Grade WC6 or A387 Grade 11.
10	Group 1.10	A182 Grade F22, A217 Grade WC9 or A387 Grade 22.
13	Group 1.13	A182 Grade F5a or A217 Grade C5.
14	Group 1.14	A182 Grade F9 or A217 Grade C12.
21	Group 2.1	A182 Grade F304 or F304H, A240 Type 304 or A351 Grade CF3 or CF8.
22	Group 2.2	A182 Grade F316 or F316H, A240 Type 316 or 317 or A351 Grade CF3M or CF8M.
23	Group 2.3	A182 Grade F304L or F316L or A240 Type 304L or 316L.
24	Group 2.4	A182 Grade F321 or F321H or A240 Type 321 or 321H.
25	Group 2.5	A182 Grade F347, F347H, F348 or F348H, A240 Type 347, 347H, 348 or 348H or A351 Grade CF8C.
26	Group 2.6	A240 Type 309S or A351 Grade CH8 or CH20.
27	Group 2.7	A182 Grade F310, A240 Type 310S or A351 Grade CK20.

5. Reduced pipe size. Reducing slip-on welding flanges and the size to which the reduction is being made are identified by a single letter (see table IV).

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TABLE IV. Code letter to reduced nominal pipe size.

Code Letter	Reduced Size (Nominal Pipe Size-Inch)
A	1/2
B	3/4
C	1
D	1-1/4
E	1-1/2
F	2
G	2-1/2
H	3
J	3-1/2
K	4
M	5
N	6
P	8
R	10
S	12
T	14
U	16
W	18
X	20

7. Military specification part number definition. The military specification sheet part number is a definitive part number which is formed by combining the Military specification sheet code number (M18180/3 for MIL-F-18180/3) the pipe size and pressure-temperature rating class code number, the flange facing code letter, the ANSI B16.5 materials group number code number and, if applicable, the reduced size code letter as shown in paragraph 1.

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(Project 4730-0706)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-F-18180/3A	FLANGES AND FLANGED FITTINGS, PIPE, STEEL, CLASSES 150, 300, 400, 600, 900, 1500, AND 2500

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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